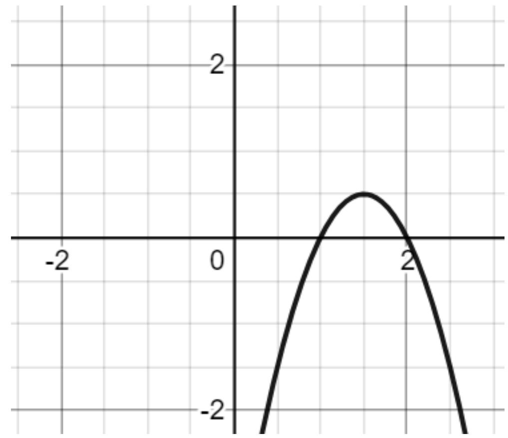
Algebra 1 Name____________________
Unit 8 Date____________________
Lesson 2 Homework Period____________________

1. As the x- values of the parabola get infinitely *larger*, the y-values of the parabola approach

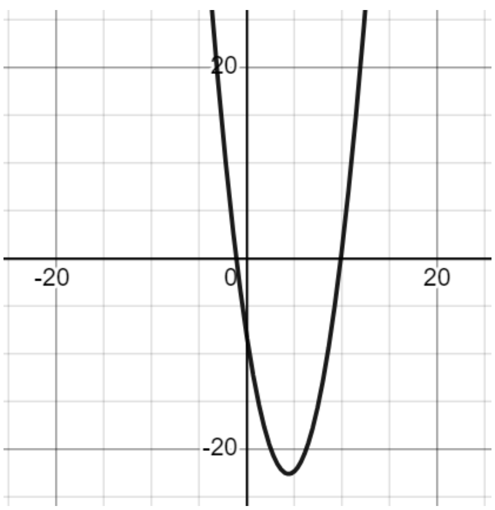
   ☐ ∞
   ☐ $-\infty$



2. As the x-values of the parabola get infinitely *smaller*, the y-values of the parabola approach

   ☐ ∞
   ☐ $-\infty$

Use the graph of the quadratic equation for questions 3 and 4.



3. Use words or symbols to describe the domain of the quadratic function.

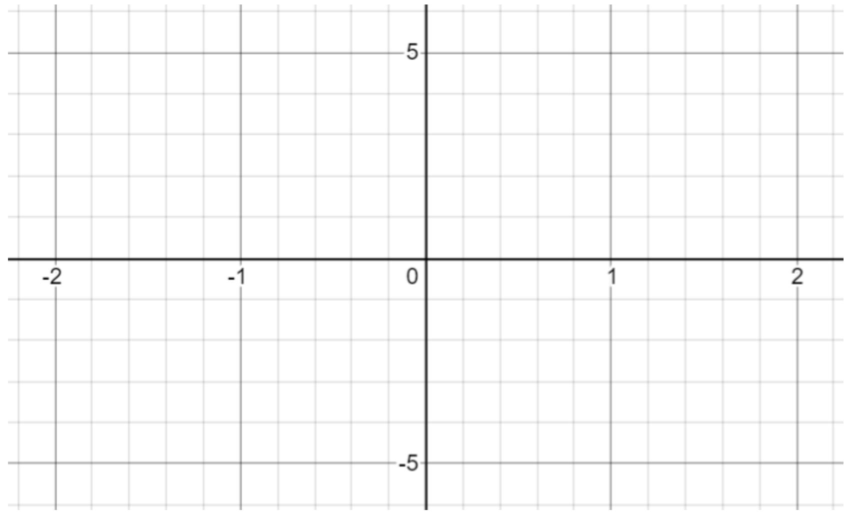
4. Use words or symbols to describe the range of the quadratic function.

5. Determine the vertex and y –intercept of a quadratic using the equation $y = -5(x-1)^2 + 3$

6. Use the table of values to determine the vertex of the parabola.

x	y
0.1	−1.05
0.2254	0
0.5	1.75
1	3
1.5	1.75
1.7746	0
2	−2

7. Use the key features you determined in question 5 and 6 to sketch the graph of the quadratic function.



8. Identify key features of the quadratic $y = -(x-3)^2 + 4$

x-intercept		vertex		domain		left end behavior	$x \to -\infty$, $y \to$ _____
x-intercept		increasing		range		right end behavior	$x \to \infty$, $y \to$ _____
y-intercept		decreasing		opens			

9. Create a table of values for the quadratic function in question 8.

x						
y						

10. Use the key features and table of values from 7 and 8 to graph the quadratic.